Milton Tran

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Education

UC Riverside — Riverside, CA
Undergraduate Student in Mechanical Engineering — *Expected Graduation: Jun 2030*

BASIS Independent Fremont Upper — Fremont, CA
High Honors — *Aug. 2022 – May 2026*

Experience

FRC Team 9470 — Fremont, CA
Mechanical Design Lead — *Jun. 2024 – May 2026*

- Led the mechanical design team through robot design, fabrication, assembly, and iteration.
- Onboarded new team members in robot design, mechanical assembly, and team workflows.
- Coordinated pit crew preparation, robot maintenance, troubleshooting, and competition repairs.

FRC Team 9202 — San Jose, CA
Technical Captain — *Dec. 2022 – May 2024*

- Led the mechanical design team through robot design, fabrication, assembly, and iteration.
- Maintained and inspected the robot's electrical systems to ensure reliable operation.
- Collaborated with a local makerspace to organize and support hands-on robotics workshops.

Build Play Learn Labs — Menlo Park, CA
STEM Instructor — *Jul. 2024 – Present*

- Instruct students in hands-on STEM activities using LEGO-based construction, 3D printing, and block-based programming.
- Introduce students to 3D printing through interactive lessons and project-based activities.
- Maintain 3D printers and classroom equipment to support safe, reliable classroom operations.

School Involvement

Highlander Racing, Riverside, CA — September 2026 – Present

- *Suspension Subteam*
- Designing suspension and steering components for the Formula SAE car using SolidWorks CAD and FEA
- Analyzing component geometry and structural performance to support suspension design and vehicle dynamics

Highlander Combat Robotics, Riverside, CA — October 2026 – Present

- *Mechanical Subteam*
- Designing mechanical components and assemblies for a plastic combat robot using CAD
- Developing lightweight, functional mechanical designs to support the robot's performance and durability

Awards

- FIRST World Championship 2026 Archimedes Division Champion, top 0.01% at international competition
- FIRST Half Moon Bay and Silicon Valley District Champion 2026
- FIRST Excellence in Engineering and Industrial Design Awards
- Presidential Volunteer Service Gold Award 2023, volunteered for 100+ hours

Technical Skills

Softwares: Onshape, Solidworks, Fusion 360, OrcaSlicer
Manufacturing and Fabrication: FDM and SLA 3D Printing, CNC Machining, Assembly
Electronics: Arduino Uno, Universal G-code Sender, Wiring
Programming: Python, Java, VS Code
Languages: Mandarin, Cantonese